CSE 312

Foundations of Computing II

Lecture 2: Combinations and Binomial Coefficients

Announcements

Homework:

- Pset1 was posted on Wednesday and is due 11:59pm next Wednesday.
  - Note the AI policy: no AI on graded work

- We will have the same pattern for the other assignments, excluding the week of the midterm and the last two assignments after Thanksgiving, which will be due on Fridays.

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Python programming on homework:

- Some problem sets will include coding problems
  - in Python (*no prior knowledge or experience required*)
  - provide a deeper understanding of how theory we discuss is used in practice
  - should be fun

Announcements

- **Problem Set 1**
  - Read the first page for how to write up your homework solutions. Don't wait until you are working on the questions to figure it out!
  - Section solutions are a good place to look for examples.
- **Resources**
  - Textbook readings can provide another perspective
  - Theorems & Definitions sheet – https://www.alextsun.com/files/defs_thms.pdf
  - Office Hours
  - EdStem discussion
- **EdStem discussion etiquette**
  - OK to publicly discuss content of the course and any confusion over topics discussed in class, but **not *solutions*** for current homework problems, or anything about current exams that have not yet been graded.
  - It is also acceptable to ask for clarifications about what current homework problems are asking and concepts behind them, just not about their solutions.

Quick counting summary from last class

- Sum rule:

  If you can choose from
  - EITHER one of n options,
  - OR one of m options with NO overlap with the previous n,

  then the number of possible outcomes of the experiment is $n + m$

- Product rule:

  In a sequential process, if there are
  - n_1 choices for the 1st step,
  - n_2 choices for the 2nd step (given the first choice), ..., and
  - n_k choices for the kth step (given the previous choices),

  then the total number of outcomes is $n_1 \times n_2 \times n_3 \times \cdots \times n_k$

- Representation of the problem is important (creative part)

Factorial

"How many ways to order elements in S, where $|S| = n$?" **Permutations**

Answer = $n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$

Definition. The **factorial function** is

$$n! = n \times (n-1) \times \cdots \times 2 \times 1$$

Note: $0! = 1$

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Theorem. (Stirling's approximation)

$$\underbrace{\sqrt{2\pi}}_{= 2.5066} \cdot n^{n+\frac{1}{2}} \cdot e^{-n} \leq n! \leq \underbrace{e}_{= 2.7183} \cdot n^{n+\frac{1}{2}} \cdot e^{-n}$$

Huge: Grows exponentially in n

Distinct Letters

"How many sequences of 5 ***distinct*** *alphabet letters from* $\{A, B, \ldots, Z\}$*?"*

E.g., AZURE, BINGO, TANGO. But not: STEVE, SARAH

Answer: $26 \times 25 \times 24 \times 23 \times 22$

In general

Aka: k-permutations

Fact. # of k-element sequences of distinct symbols from an n-element set is

$$P(n,k) = n \times (n-1) \times \cdots \times (n-k+1) = \frac{n!}{(n-k)!}$$

Distinct Letters

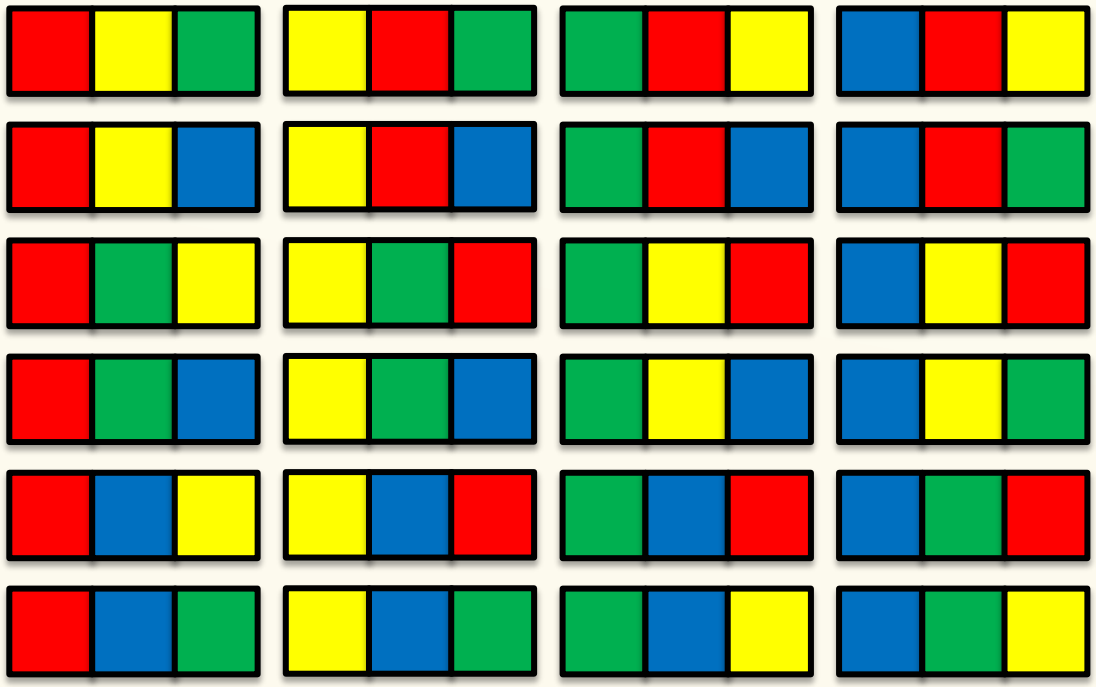
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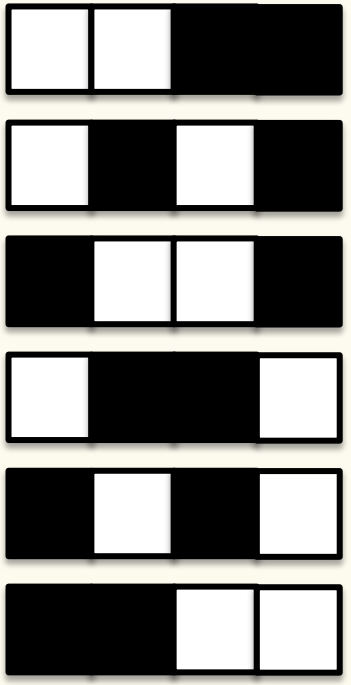
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Answer: $26 \times 25 \times 24 \times 23 \times 22 = \frac{26!}{(26-5)!} = \frac{26!}{21!}$

Today: More Counting

- Permutations and Combinations





Number of Subsets

"How many size-5 ***subsets*** *of* $\{A, B, \ldots, Z\}$*?"*

E.g., {A,Z,U,R,E}, {B,I,N,G,O}, {T,A,N,G,O}. But not: {S,T,E,V}, {S,A,R,H},…

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Difference from k-permutations: NO ORDER

Different sequences: TANGO, OGNAT, ATNGO, NATGO, ONATG …

Same set: {T,A,N,G,O}, {O,G,N,A,T}, {A,T,N,G,O}, {N,A,T,G,O}, {O,N,A,T,G}… …

Number of Subsets – Idea

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Consider a sequential process:

1. Choose a subset $S \subseteq \{A, B, \dots, Z\}$ of size $|S| = 5$
   e.g. $S = \{A, G, N, O, T\}$

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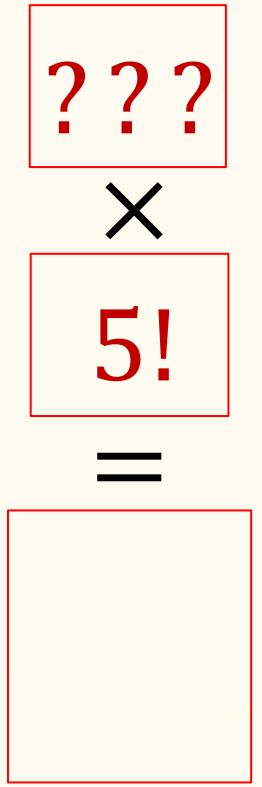
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   5!

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$$???$$
$$\times$$
$$5!$$
$$=$$
$$\frac{26!}{21!}$$

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$$??? = \frac{26!}{21!\,5!} = 65780$$

Number of Subsets – Binomial Coefficient

Fact. The number of subsets of size k of a set of size n is

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

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Binomial coefficient (verbalized as "n choose k")

Notation: $\binom{S}{k}$ = set of all k-element subsets of S.

[also called **combinations**]

$$\left|\binom{S}{k}\right| = \binom{|S|}{k}$$

Symmetry in Binomial Coefficients

Fact. $\binom{n}{k} = \binom{n}{n-k}$

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Proof. $\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k}$

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Why??



Symmetry in Binomial Coefficients – A different proof

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Two equivalent ways to choose k out of n objects (unordered)

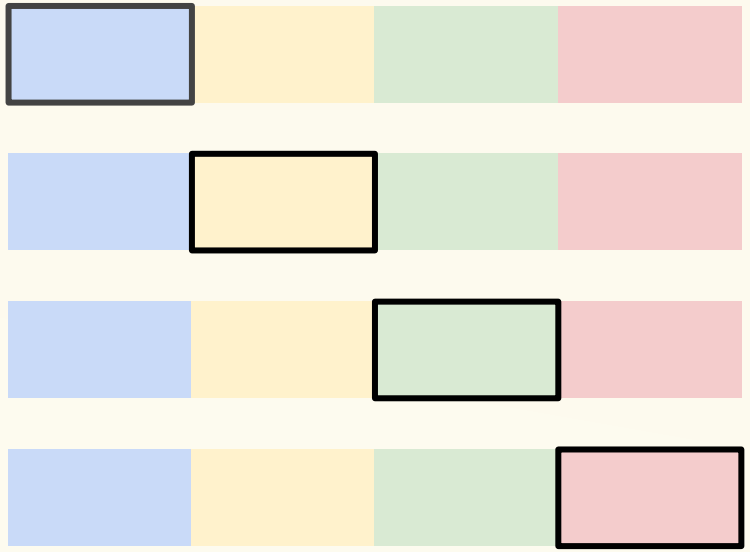
1. Choose which k elements are included
2. Choose which $n - k$ elements are excluded

Symmetry in Binomial Coefficients – A different proof

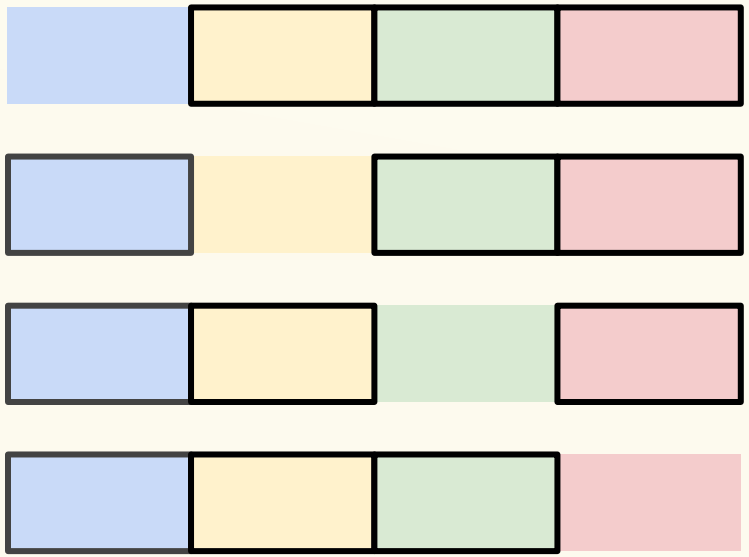
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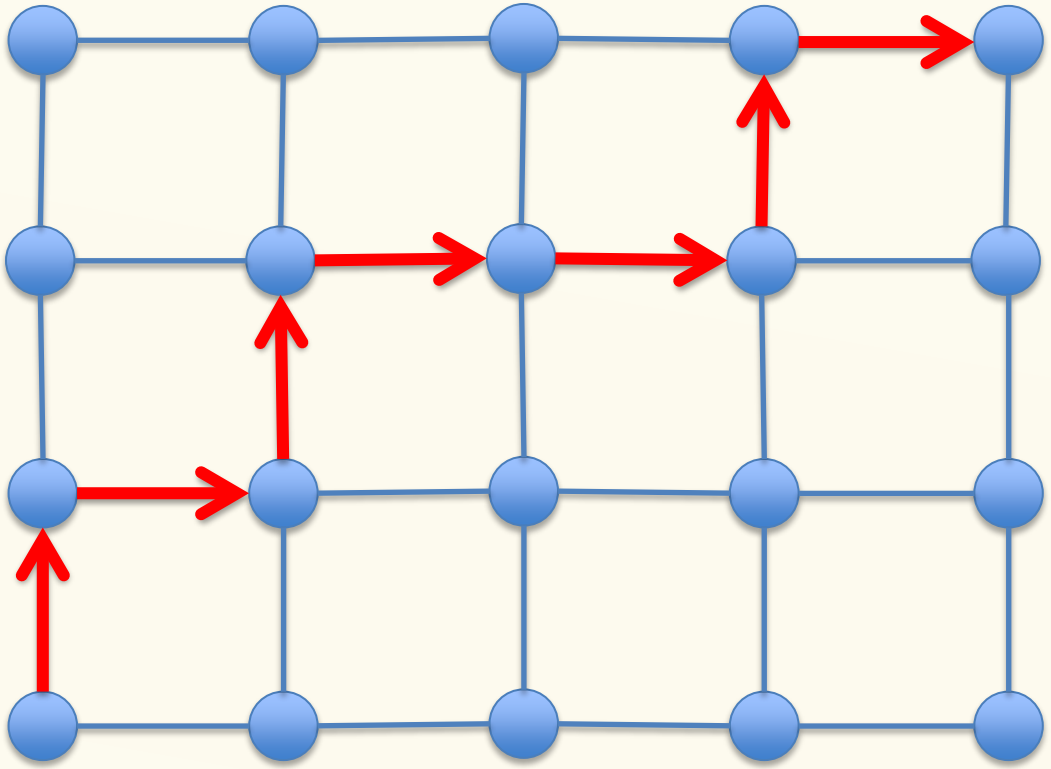
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$$\binom{4}{1} = 4 = \binom{4}{3}$$



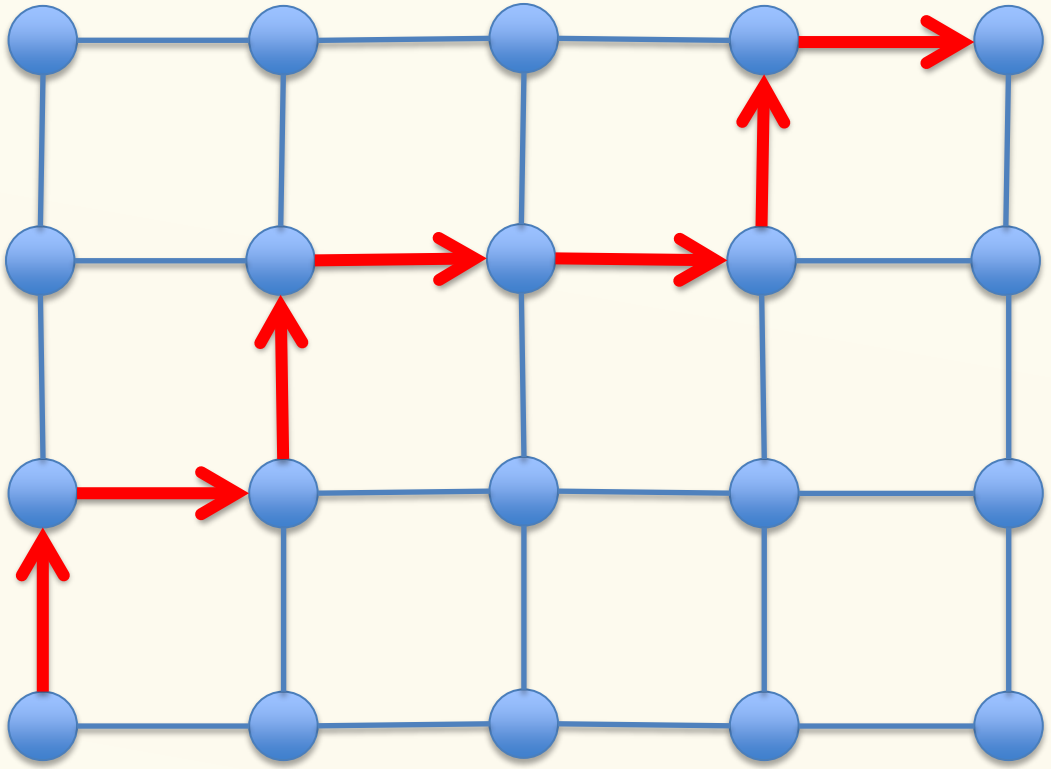
Example – Counting Paths





"How many shortest paths from Gates to Starbucks?"

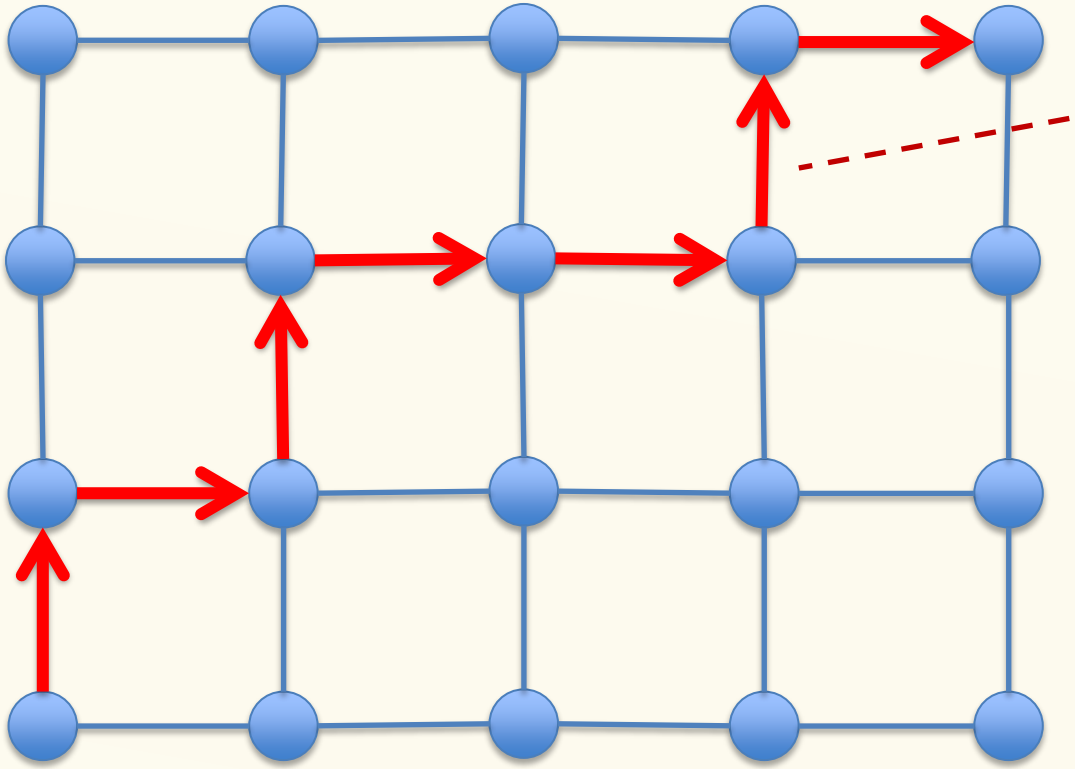
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How do we represent a shortest path?

Example – Counting Paths

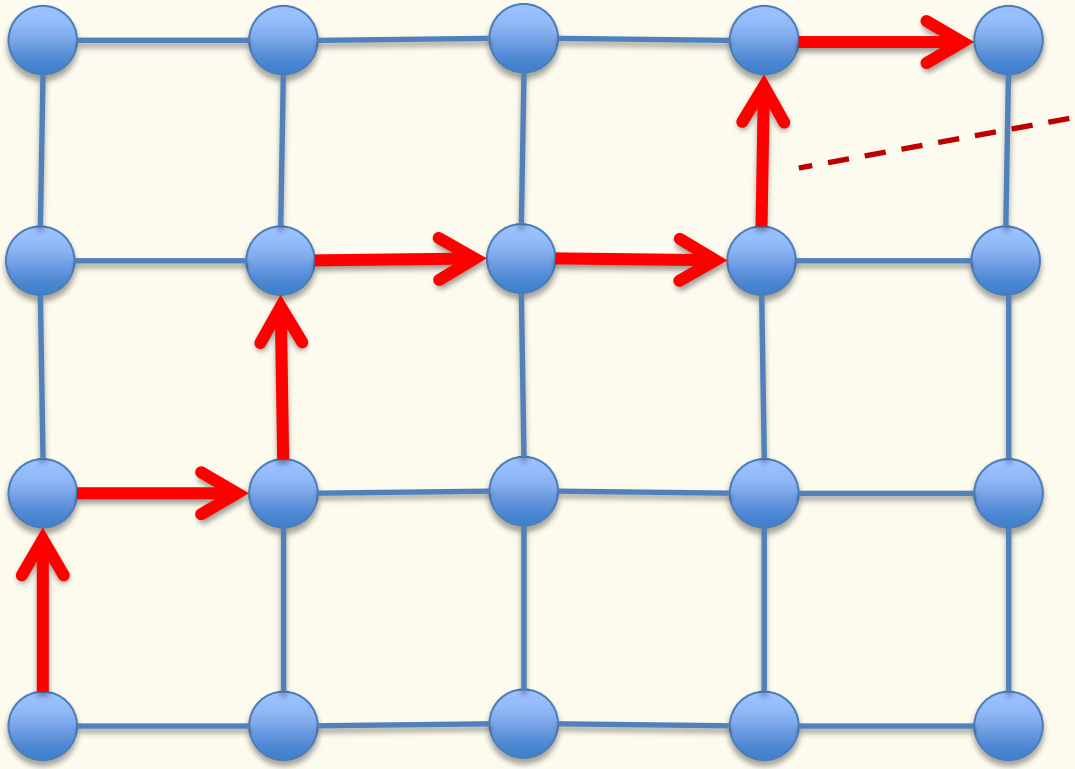


$$\text{Path} \in \{\uparrow, \rightarrow\}^7$$

$$(\uparrow, \rightarrow, \uparrow, \rightarrow, \rightarrow, \uparrow, \rightarrow)$$



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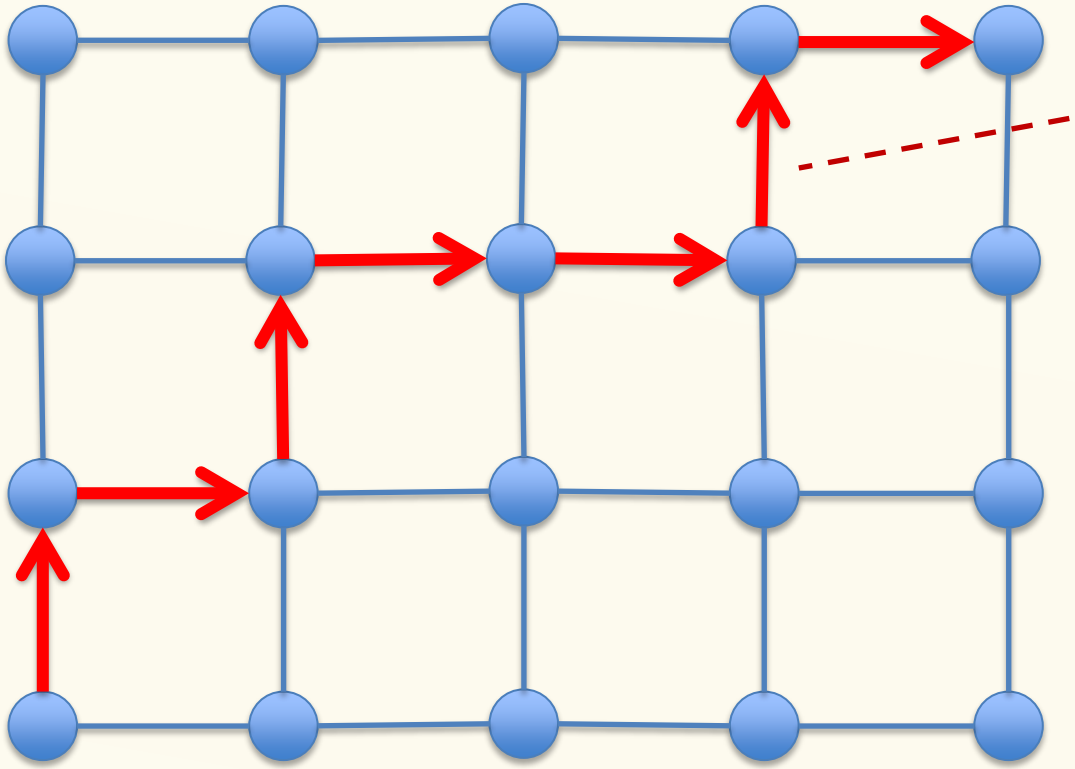
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↑'s =3, # →'s=4



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$(\uparrow, \rightarrow, \uparrow, \rightarrow, \rightarrow, \uparrow, \rightarrow)$

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Poll: # of shortest paths?

A. 2^7

B. $\frac{7!}{4!}$

C. $\binom{7}{4} = \frac{7!}{4!3!}$

D. *No idea*



Example – Sum of integers

"How many solutions $(x_1, \dots, x_k)$ *such that* $x_1, \dots, x_k \geq 0$ *and* $\sum_{i=1}^{k} x_i = n$*?"*

Example: $k = 3, n = 5$

$(0,0,5), (5,0,0), (1,0,4), (2,1,2), (3,1,1), (2,3,0), \dots$

Hint: we can represent each solution as a binary string.

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Clever representation of solutions

$(3,1,1)$ → $1\ 1\ 1\ 0\ 1\ 0\ 1$

$(2,1,2)$ → $1\ 1\ 0\ 1\ 0\ 1\ 1$

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sols = # strings from $\{0,1\}^7$ w/ exactly two 0s

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sols = # strings from $\{0,1\}^{n+k-1}$ with $k-1$ 0s

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$$= \binom{n+k-1}{k-1}$$

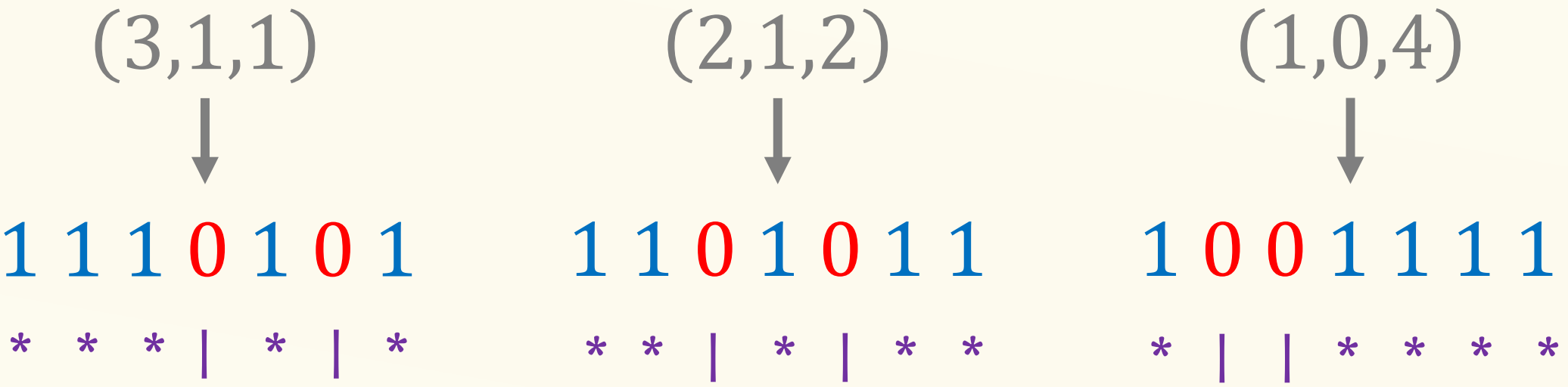
After a change in representation, the problem magically reduces to counting combinations.

Example – Sum of integers

Example: $k = 3, n = 5$

sols = # strings from $\{0,1\}^7$ w/ exactly two 0s $= \binom{7}{2} = 21$

Clever representation of solutions



Also called the "stars and bars" method

*aka. “stars and bars method”

More general counting using binary encoding*

The number of ways to distribute n indistinguishable balls into k distinguishable bins is

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$$

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The number of ways to distribute n indistinguishable balls into k distinguishable bins is

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Example with k non-negative integers summing to n:
bins are the k integers, balls are the n 1’s that add to n.

Coins

How many ways can you distribute 32 identical coins among Alex, Barbara, Charlie, Dana, and Eve?



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1. Identify balls
2. Identify bins



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2. Identify bins

$$\binom{32+5-1}{5-1}$$

A mixed example – Word Permutations (aka Anagrams)

"How many ways to re-arrange the letters in the word SEATTLE?

STALEET, TEALEST, LASTTEE, …

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No! e.g., swapping two T's also leads to *SEATTLE*
swapping two E's also leads to *SEATTLE*

Counted as separate permutations, but they lead to the same word.

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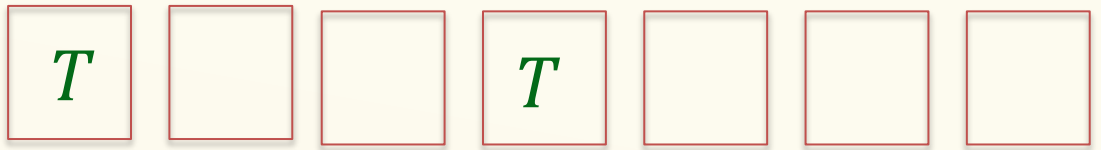
Locations of T's

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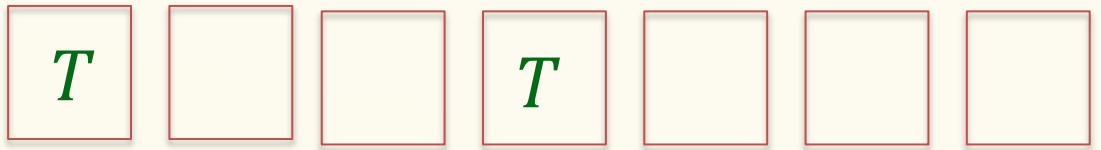
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$$\binom{7}{2} \times \binom{5}{2}$$

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Locations of E's



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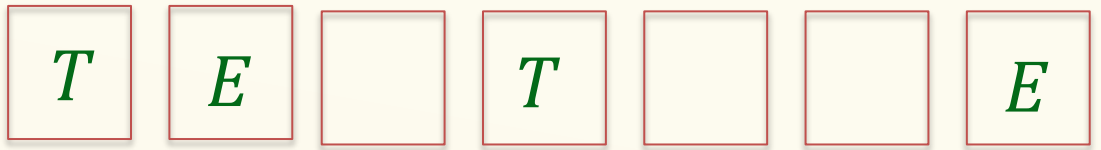
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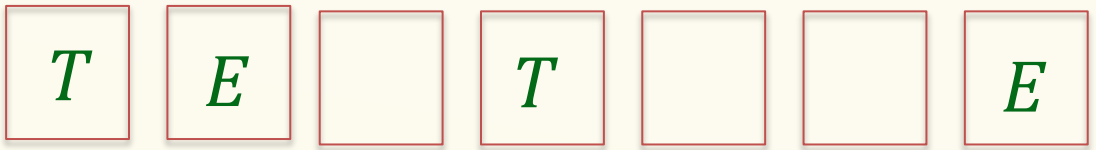
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Location of S

$$\binom{7}{2} \times \binom{5}{2} \times 3$$

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Locations of E's



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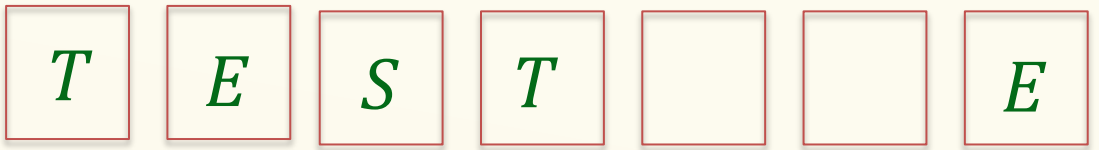
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Location of S

Locations of T's

Locations of E's



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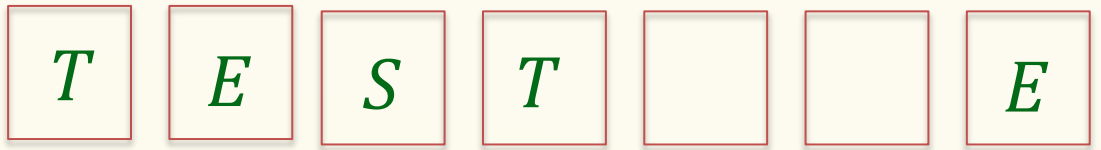
$$\binom{7}{2} \times \binom{5}{2} \times 3 \times 2$$

Location of S

Location of L

Locations of E's

Locations of T's



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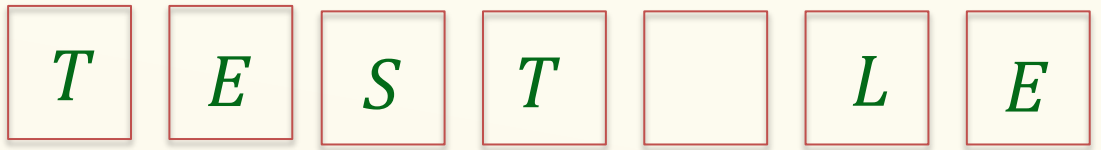
Location of S

$$\binom{7}{2} \times \binom{5}{2} \times 3 \times 2$$

Location of L

Locations of T's

Locations of E's



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Location of S

Location of A

Location of L

Locations of E's

Locations of T's

T	E	S	T	A	L	E

Another way to look at SEATTLE

"How many ways to re-arrange the letters in the word SEATTLE?

STALEET, TEALEST, LASTTEE, …

$$\binom{7}{2} \times \binom{5}{2} \times 3 \times 2 \times 1 = \frac{7!}{2!\,5!} \times \frac{5!}{2!\,3!} \times 3!$$

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Another interpretation:

Arrange the 7 letters as if they were distinct. Then divide by 2! to account for 2 duplicate T's, and divide by 2! again for 2 duplicate E's.

More generally…

How many ways can you arrange the letters in “Godoggy”?



$n = 7$ Letters, $k = 4$ Types {G, O, D, Y}

$n_1 = 3, n_2 = 2, n_3 = 1, n_4 = 1$

$$\frac{7!}{3!2!1!1!} = \binom{7}{3,2,1,1}$$

Multinomial coefficients

Multinomial Coefficients

If we have k types of objects (n total), with n_1 of the first type, n_2 of the second, …, and n_k of the kth, then the number of orderings possible is

$$\binom{n}{n_1, n_2, \cdots, n_k} = \frac{n!}{n_1!\, n_2! \cdots n_k!}$$

Binomial Coefficients – Many interesting and useful properties

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

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Follows from Binomial Theorem

Binomial Coefficients – Many interesting and useful properties

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Fact. $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ — **Follows from Binomial Theorem**

Fact. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ — **Pascal's Identity (Next lecture)**

Binomial Theorem: Idea

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$$(x + y)^2 = (x + y)(x + y)$$

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$$(x+y)^2 = (x+y)(x+y)$$
$$= xx + xy + yx + yy$$

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$$
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$$

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Binomial Theorem: Idea

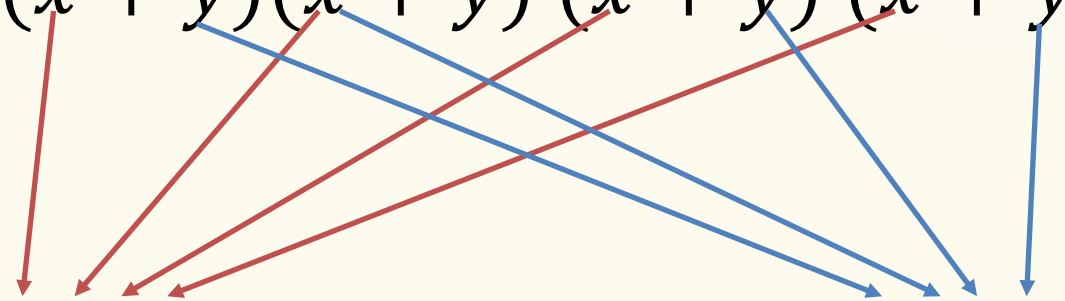
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Poll: What is the coefficient for xy^3?

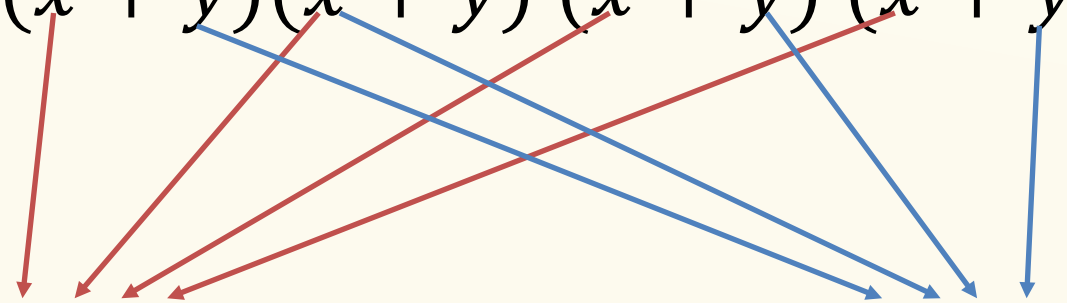
A. 4

B. $\binom{4}{1}$

C. $\binom{4}{3}$

D. 3

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Binomial Theorem: Idea

$$(x+y)^n = (x+y)\ldots(x+y)$$

Each term is of the form $x^k y^{n-k}$, since each term is made by multiplying exactly n variables, either x or y, one from each copy of $(x+y)$

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Theorem. Let $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$ a positive integer. Then,

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Apply with $x = y = 1$

Many properties of sums of binomial coefficients can be found by plugging in different values of x and y in the Binomial Theorem.

Corollary.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Quick Summary

- k-sequences: How many length k sequences over alphabet of size n?
  - Product rule $\rightarrow n^k$
- k-permutations: How many length k sequences over alphabet of size n, without repetition?
  - Permutation $\rightarrow \frac{n!}{(n-k)!}$
- k-combinations: How many size k subsets of a set of size n (without repetition and without order)?
  - Combination $\rightarrow \binom{n}{k} = \frac{n!}{k!(n-k)!}$